

AI vs Human Text Detection

Exploring how text-mining methods distinguish between machine-generated and human-written content

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The Challenge of Authenticity

Generative AI tools like ChatGPT, Bard, and Claude have become central to research, learning, and writing. As large language models grow more sophisticated, distinguishing between human and machine-generated text has become critical. This distinction matters greatly in domains where authenticity is essential for trust, accountability, and integrity.



Why Detection Matters



Education

Academic institutions worry about dishonesty as students increasingly use AI for assignments.



News Media

Organizations face synthetic misinformation spreading at unprecedented speed and scale.



Research

Scientific publishers must safeguard integrity against fabricated submissions.



Legal & Medical

Sectors where accuracy and authorial responsibility are essential now confront convincing AI documents.

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The Detection Arms Race

> As detection methods improve, generation models also evolve and develop techniques to evade detection.

Many research papers focus on identifying AI-generated writing through signals like em dashes, vague phrasing, or overly sophisticated language. However, far less attention has been given to detecting human writing.

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Research Foundation

1

Ippolito et al. (2020)

Machine-generated texts that successfully mimic human writing are the most difficult for models to classify correctly.

2

Mitchell et al. (2023)

Developed DetectGPT, which detects AI text by analyzing probability distribution curvature in a zero-shot setting.

3

Hayawi et al. (2023)

Demonstrated that narrative and opinion writing are significantly more challenging to classify than other genres.

4

Jakesch et al. (2023)

Found that humans rely on misleading indicators like emotional tone when identifying AI text.

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Dataset & Methodology

Data Source

Kaggle "AI vs Human Text Detection" collection with nearly half a million labeled samples.

Sample Size

Stratified subset of **1,000 total samples** split into training and test sets.

Topics Covered

- Politics
- Environment
- Transportation
- Personal reflections
- Social issues

Authentic Variation

Original grammar, punctuation, and tone preserved to expose classifier to real-world imperfections.

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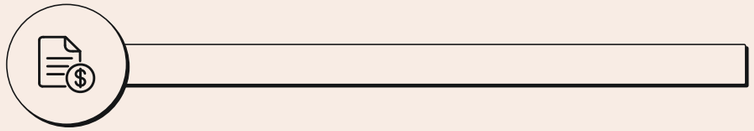
90% Accuracy

The classifier's best performance reached approximately 0.9 accuracy—a strong result for a dataset with mixed topics and natural variation.

This demonstrates that even relatively simple models can learn meaningful distinctions between human and AI text when trained on sufficiently representative examples.

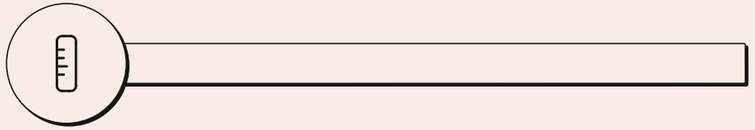


Improving Through Error Analysis



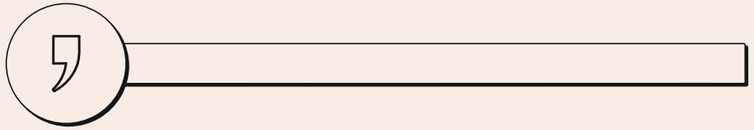
Remove Noise

Removed all quotation marks from the dataset, as embedded dialogues introduced noise without contributing to classification.



Line Length Variation

Added average line length as a measure. Human writers vary sentence length naturally, while AI produces more uniform structures.



Punctuation Patterns

Incorporated punctuation amount. Human writing shows inconsistent punctuation, while AI follows predictable patterns.

These adjustments improved and stabilized the classifier's performance at its best accuracy value of 0.9

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Top Features for Human Detection

because

Suggests conversational or causal reasoning humans use frequently

getting

Appears naturally in everyday writing and speech patterns

being

Common in human expression and natural language flow

says

Reflects conversational attribution typical of human writing

rsb

Uncommon token that appeared more often in human text

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Key Takeaways

Strong Performance

Achieved 90% accuracy in distinguishing human from AI text across diverse topics and writing styles.

Human Patterns Matter

Features like conversational reasoning, natural variation, and inconsistent punctuation are strong human indicators.

Ongoing Challenge

As AI models evolve, detection methods must continuously adapt to maintain effectiveness.

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